

CBSE Test Paper 01
Chapter 3 Motion in A Straight Line

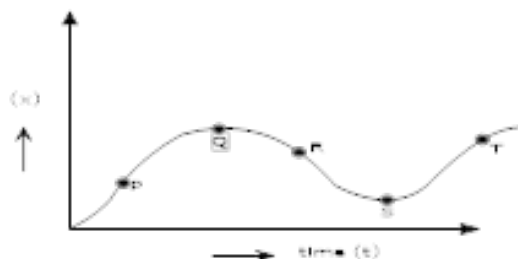
1. For motion with uniform acceleration, x-t graph is **1**
 - a. a parabola
 - b. a hyperbola
 - c. a straight line
 - d. a cubic

 2. A jet lands on an aircraft carrier at 63 m/s. What is its acceleration in m/s^2 if it stops in 2.0 s? **1**
 - a. -35
 - b. 34
 - c. -31
 - d. -33

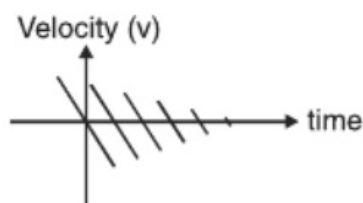
 3. A ball starts from rest and accelerates at 0.500 m/s^2 while moving down an inclined plane 9.00 m long. When it reaches the bottom, the ball rolls up another plane, where, after moving 15.0 m, it comes to rest. What is the speed of the ball in m/s at the bottom of the first plane? **1**
 - a. 3.0
 - b. 1.00
 - c. 2.00
 - d. 4.00

 4. A golf ball is released from rest from the top of a very tall building. Calculate the position in m of the ball after 2.00 s **1**
 - a. -32.1
 - b. -19.6
 - c. -22.2
 - d. -20.9
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5. A drag racer starts her car from rest and accelerates at 10.0 m/s^2 for the entire distance of 400 m .How long did it take the race car to travel this distance in s? **1**
- 9.01
 - 8.33
 - 10.2
 - 8.94
6. Can a body have zero velocity and still be accelerating? If yes, give any situation. **1**
7. Figure shows a displacement- time graph. Comment on the sign of velocities at point P, Q, R, S and T. **1**

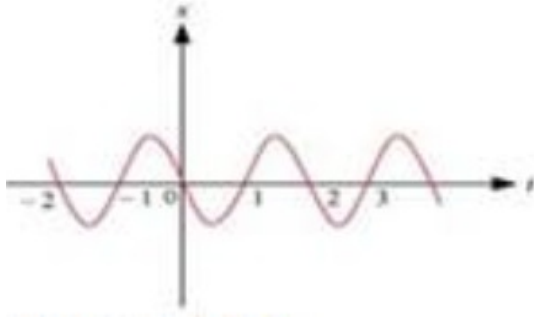


8. Suggest a suitable physical situation for the following graph. **1**



9. A person travels along a straight road for the first half with a velocity v_1 and the second half with velocity v_2 . What is the mean velocity of the person for the whole journey? **2**
10. Write the characteristics of displacement. **2**
11. Sameer went on his bike from Delhi to Gurgaon at a speed of 60km/hr and came back at a speed of 40km/hr. what is his average speed for entire journey. **2**
12. A particle moving in a straight line covers half the distance with a speed of 3 m/s. The other half of distance is covered in two equal intervals of time with speeds of 4.5 m/s and 7.5 m/s, respectively. Find the average speed of the particle during this motion. **3**

13. Figure gives the x - t plot of a particle executing one-dimensional simple harmonic motion. Give the signs of position, velocity and acceleration variables of the particle at $t = 0.3$ s, 1.2 s, -1.2 s. **3**



14. A particle is moving along a straight line and its position is given by the relation $x = (t^3 - 6t^2 - 15t + 40)$ m. **3**
Find
- The time at which velocity is zero.
 - Position & Displacement at this point
 - Acceleration for the particle at the point.
15. Two particles move along x - axis. The position of particle 1 is given by $x = 6.00t^2 + 3.00t + 2.00$ (in metre and in seconds); acceleration of particle 2 is given by $a = -8.00t$ (in m/s^2 and seconds) and at $t = 0$, its velocity is 20 m/s. When the velocities of the particles match, find their velocities. **5**